



5.5 Integration and areas under curves

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Integration is the reverse of differentiation: $\int x^n dx = x^{(n+1)}/(n+1) + C$ ($n \neq -1$), plus standard results like $\int e^x dx = e^x + C$, $\int 1/x dx = \ln|x| + C$. The definite integral $\int [a \text{ to } b] f(x) dx$ gives the signed area between the curve and the x-axis from $x=a$ to $x=b$ (area below the axis counts as negative). The area between two curves $y=f(x)$ and $y=g(x)$ ($f \geq g$ on $[a,b]$) is $\int [a \text{ to } b] (f(x)-g(x)) dx$.

Worked example: follow the reasoning

Find the area enclosed between $y=4-x^2$ and the x-axis.

x-intercepts: $4-x^2=0 \rightarrow x=\pm 2$.

Area = $\int [-2 \text{ to } 2] (4-x^2) dx = [4x-x^3/3]$ from -2 to 2 = $(8-8/3)-(-8+8/3) = (16/3)-(-16/3) = 32/3$.

Quick check · multiple choice

What is $\int [0 \text{ to } 2] 3x^2 dx$?

- A 4
- B 6
- C 8
- D 12

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Consider the curves $y=x^2+1$ and $y=2x+4$.
 - (a) Find the x-coordinates of the points of intersection.
 - (b) Find the area enclosed between the two curves.

[7]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $x^2+1=2x+4 \rightarrow x^2-2x-3=0 \rightarrow (x-3)(x+1)=0 \rightarrow x=3$ or $x=-1$.

(b) On $[-1,3]$, the line $2x+4$ is above x^2+1 (check at $x=0$: line= $4 > 1$ =curve). Area = $\int_{-1}^3 [(2x+4)-(x^2+1)]dx = \int_{-1}^3 (-x^2+2x+3)dx = [-x^3/3+x^2+3x]$ from -1 to 3 .

At $x=3$: $-9+9+9=9$. At $x=-1$: $1/3+1-3=-5/3$.

Area = $9-(-5/3) = 9+5/3 = 32/3$.

Quick check answer

C. $\int 3x^2 dx = x^3$, evaluated: $2^3-0^3=8$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.